

Distributed Task Execution Platform Using Chord DHT

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Abstract—Traditional distributed task execution platforms rely on centralized coordinators, which introduce single points of failure and bottlenecks in fault-prone environments. We present a fully decentralized job execution platform built on the Chord Distributed Hash Table (DHT) protocol, augmented with intelligent task placement and automated failure recovery. The system distributes task metadata and worker assignments across two logically distinct hash tables stored on the same Chord ring, with $k=3$ replication for durability. An AI-driven placement agent scores workers based on load, latency, and historical success rate to route tasks intelligently, while a recovery agent detects worker crashes within seconds and automatically reassigns tasks to surviving workers. We implement seven phases: Chord ring construction, conversational AI interface, atomic two-write persistence, worker execution, natural-language status reporting, intelligent placement, and intelligent failure recovery. Evaluation on a multi-node ring demonstrates $O(\log N)$ lookup performance, automatic task recovery after worker failure with zero user intervention, and successful end-to-end task execution through a conversational interface. The system eliminates the central coordinator entirely while providing self-healing capabilities and load-aware task routing.

Index Terms—Chord protocol, distributed hash tables, fault tolerance, intelligent placement, failure recovery, peer-to-peer systems, distributed job execution

I. INTRODUCTION

Modern distributed computing environments demand job execution platforms that can scale horizontally, tolerate failures, and operate without single points of failure. Traditional systems such as Apache Celery, Apache YARN, and Kubernetes rely on centralized coordinators or master nodes for task routing and scheduling decisions. While effective at data-center scale, this architecture introduces two fundamental limitations: (i) the coordinator represents a single point of failure whose loss disrupts the entire system, and (ii) scheduling decisions funnel through a central bottleneck, limiting horizontal scalability and increasing operational complexity.

Peer-to-peer distributed hash tables provide an alternative coordination substrate. The Chord protocol, introduced by Stoica et al., arranges nodes in a consistent-hash ring and resolves key ownership in $O(\log N)$ hops via finger tables.

While Chord has been studied extensively as a lookup mechanism, its application as a complete job execution platform with intelligent task placement and automated failure recovery remains underexplored.

This paper presents a distributed job execution platform that uses the Chord ring as the storage substrate for both task records and worker assignments, while layering intelligent decision-making on top through dedicated placement and recovery agents. Our design contributions are as follows:

- **Dual-DHT storage model.** Two logically distinct hash tables are maintained on the same Chord ring: HT1 stores task records keyed by `result:{task_id}`, and HT2 stores worker assignments keyed by `worker:{worker_id}`. Both tables use $k=3$ successor-chain replication for fault tolerance.
- **Atomic two-write protocol.** Task submission requires consistent writes to both HT1 and HT2. The system implements idempotency checks, exponential-backoff retries, and rollback on partial failure to ensure either both writes succeed or neither persists.
- **Intelligent placement agent.** A heuristic scoring function evaluates each live worker using weighted load, latency, and historical success rate metrics. Tasks are routed to the highest-scoring worker, replacing naive round-robin assignment with load-aware placement.
- **Automated failure recovery.** A recovery agent continuously monitors worker heartbeats and detects three failure modes: worker crashes, task timeouts, and execution failures. For each detected failure, the agent selects a recovery path (retry on different worker, wait and retry, or give up) based on attempt count, failure type, and cluster state.
- **Conversational AI interface.** A natural-language agent enables users to submit tasks, check status, and receive recovery notifications through conversational interaction rather than direct API calls.

We evaluate our system on a multi-node ring and demonstrate: (i) $O(\log N)$ hop count distribution validating the Chord routing efficiency, (ii) automatic task recovery within 10 seconds of worker failure, and (iii) end-to-end task execution

through conversational submission with intelligent worker selection.

The remainder of this paper is organized as follows. Section II surveys related work in distributed coordination and DHT-based systems. Section III details the seven-phase system design. Section IV describes the implementation and presents experimental results. Section V discusses limitations and future work. Section VI concludes.

II. RELATED WORK

We survey prior work across three areas: centralized job scheduling, peer-to-peer DHT protocols, and distributed coordination systems.

A. Centralized Job Scheduling Platforms

The dominant paradigm in distributed computing uses a two-tier architecture with a centralized resource manager and worker nodes. **Apache Hadoop YARN** [1] separates resource management from job lifecycle through a ResourceManager and NodeManagers, with all decisions flowing through the ResourceManager. **Apache Mesos** [3] uses a master that issues resource offers to frameworks. **Kubernetes** [2] centralizes scheduling in *kube-scheduler* and maintains cluster state in etcd [5]; while the control plane is replicated, all scheduling decisions require quorum writes to etcd.

HashiCorp Nomad [4] uses Raft-based consensus, reducing single-node risk but still requiring a designated set of coordinator nodes. **Apache Celery** is widely used for asynchronous task queues but depends entirely on external message brokers such as RabbitMQ or Redis, which themselves become single points of failure.

In contrast, our system places no coordinator requirement on the cluster. Every node is structurally identical; any node can accept and route a task request, and ring membership changes are handled by the Chord maintenance protocol itself.

B. Peer-to-Peer DHT Protocols

Distributed Hash Tables provide scalable, decentralized key-value lookup without a central directory. The canonical structured overlay protocols are Chord [6], Kademlia [7], Pastry [8], and CAN [9].

Chord [6] assigns each node and key an m -bit identifier via SHA-1 and organizes nodes in a clockwise ring. A finger table of m entries allows any node to route lookups in $O(\log N)$ hops. Ring maintenance protocols enable self-healing under node churn. Our system implements Chord directly as its routing core.

Kademlia [7] uses XOR distance and routes through k -buckets. It is widely deployed in BitTorrent and IPFS but lacks the clean ring structure that makes Chord’s successor chain a natural replication target.

Pastry [8] routes based on shared prefix length and uses a leaf set for replication. **CAN** [9] partitions a d -dimensional Cartesian coordinate space.

None of these protocols embed job execution semantics, intelligent placement, or automated failure recovery in the

TABLE I
COMPARISON OF DISTRIBUTED JOB EXECUTION APPROACHES

System	No coord.	DHT routing	Job exec.	Auto recovery	Load-aware
YARN / Mesos / K8s	×	×	✓	~	✓
Celery / RabbitMQ	×	×	✓	×	×
DynamoDB / Cassandra	~	~	×	×	×
Chord / Kademlia	✓	✓	×	×	×
CFS / PAST	✓	✓	×	~	×
Our System	✓	✓	✓	✓	✓

~ Partial support.

overlay layer. Our system builds these concerns on top of Chord while keeping the routing layer unchanged.

C. DHT-Based Storage Systems

Amazon DynamoDB [10] uses consistent hashing to partition data and replicates each key to N successive nodes. Reads and writes use a sloppy quorum with hinted handoff. **Apache Cassandra** [11] also uses a token ring with configurable replication factor. Both are storage systems and do not provide job execution abstractions.

Chord-based file systems such as CFS [12] demonstrate that successor-chain replication is feasible under churn, but apply it to file blocks rather than task metadata with execution semantics.

D. Positioning

Table I summarizes the positioning of our system relative to representative prior work.

To our knowledge, no prior system combines Chord DHT routing as the exclusive control plane for both task metadata and worker assignments with intelligent placement, automated failure recovery, and conversational task submission.

III. SYSTEM DESIGN

The system is organized in seven layers (Fig. 1), each corresponding to a development phase: a Chord ring layer, an AI agent layer, a storage layer with two distributed hash tables, a worker execution layer, a status reporting layer, an intelligent placement layer, and a failure recovery layer.

A. Phase 1: Chord Ring Layer

1) *Node Identifiers and Key Space*: Each node n is assigned an m -bit identifier $id(n) = \text{SHA-1}(\text{address}(n)) \bmod 2^m$. Our implementation uses $m = 8$ (a 256-position ring), sufficient for testing while preserving full Chord routing semantics. Keys are hashed identically.

2) *Finger Table and Lookup*: Each node maintains a finger table of m entries. Entry i stores a pointer to the first node at or clockwise of position $(n + 2^i) \bmod 2^m$. The `find_successor` operation (Algorithm 1) routes any key to its responsible node in $O(\log N)$ hops via finger table traversal.

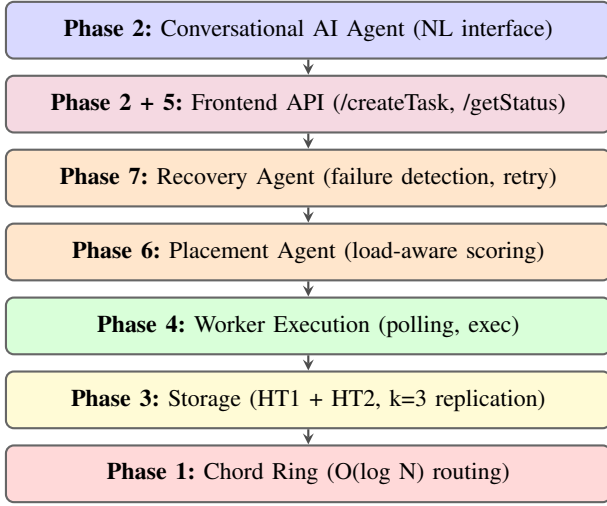


Fig. 1. Seven-phase layered architecture. Phase 2 provides the conversational interface and REST API, Phase 5 enhances status reporting on the same API layer. Higher layers depend on lower layers; the Chord ring provides the routing substrate.

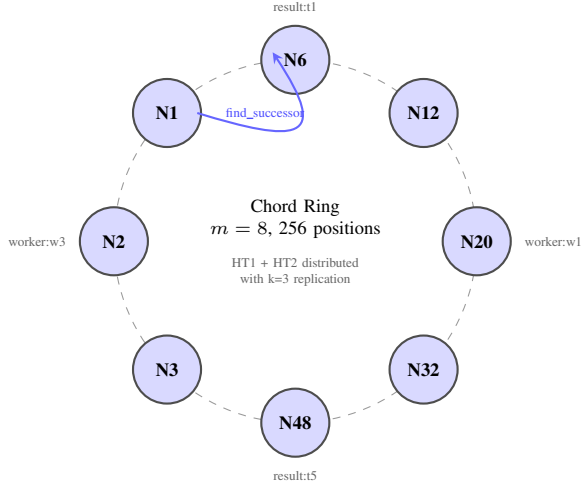


Fig. 2. Chord ring with 8 nodes. Each task or worker key hashes to a position; the responsible node owns the primary copy, with replicas on the next $k-1$ successors clockwise.

Algorithm 1: $\text{find_successor}(n, k)$

Input: Node n , target key k
Output: Successor node reference

```

1 if  $k \in (id(n), id(succ(n))]$  then
2   | return  $succ(n)$ 
3 end
4  $n' \leftarrow \text{closest\_preceding\_node}(n, k)$ 
5 if  $id(n') = id(n)$  then
6   | return  $succ(n)$ 
7 end
8 return RPC  $n'.\text{find\_successor}(k)$ 

```

3) *Ring Maintenance*: Three background daemons run on each node: `stabilize` corrects successor pointers when new nodes join, `fix_fingers` periodically refreshes finger table

Algorithm 2: Atomic Task Submission

Input: Task ID T , task details D , optional worker W
Output: HTTP response with task acceptance status

```

1 if  $HT1.get(T)$  exists then
2   | return 200, existing record           // idempotent
3 end
4 if  $W$  not provided then
5   |  $W \leftarrow \text{placement\_agent.select}(D)$ 
6 end
7  $R \leftarrow \text{build\_result\_details}(T, D, W)$ 
8 if not  $\text{retry}(HT1.put(T, R), \text{timeout}=1500\text{ms}, \text{max}=3)$  then
9   | return 503, "HT1 write failed"
10 end
11 if not  $\text{retry}(HT2.append(W, T), \text{timeout}=1500\text{ms}, \text{max}=3)$ 
    then
12   |  $HT1.delete(T)$                                // rollback
13   | return 503, "HT2 write failed"
14 end
15 return 201, "Task accepted"

```

entries, and `check_predecessor` clears stale predecessor pointers when ping fails. Together they enable the ring to self-heal under membership changes.

B. Phase 2: AI Agent and Frontend Layer

A conversational agent provides a natural-language interface for task submission and status queries. The agent translates user intent (e.g., “run my backup script”) into API calls and translates API responses into friendly status messages. The frontend exposes endpoints for task creation (`/createTask`), status queries (`/getStatus/{id}`), worker heartbeats (`/workers/heartbeat`), and live worker listing (`/workers/live`). A worker registry tracks active workers via 5-second heartbeat intervals with a 10-second timeout.

C. Phase 3: Dual-DHT Storage with Atomic Writes

The storage layer maintains two logically distinct hash tables on the same Chord ring:

- **HT1: Task Registry.** Keys of form `result:{task_id}` map to task records containing `task_type`, `script`, `path`, `status`, `worker_id`, `result`, `attempt_count`, `recovery_history`, and timestamps.
- **HT2: Worker Queues.** Keys of form `worker:{worker_id}` map to ordered lists of `task_id` values assigned to that worker.

Both tables use $k=3$ successor-chain replication with quorum-based durability. The write path for HT2 acquires a per-key lock on the primary to serialize concurrent appends to the same worker’s task list, preventing lost updates.

1) *Atomic Two-Write Protocol*: Task submission requires consistent writes to both tables. Algorithm 2 describes the protocol with idempotency, retries, and rollback.

Retries use exponential backoff with initial delay 100ms doubling each attempt. The total operation is bounded by a 5-second timeout.

D. Phase 4: Worker Execution Layer

Each worker runs two concurrent loops: a heartbeat loop that posts to `/workers/heartbeat` every 5 seconds, and a polling loop that reads its assigned tasks from HT2 every 2 seconds. For each new `task_id` not yet seen, the worker fetches the full task record from HT1, executes the script or binary via subprocess invocation with a 60-second timeout, and writes the result back to HT1 with the new status (SUCCESS or FAILURE) and captured output.

On restart, the worker reads HT2 and HT1 to identify any tasks left in PENDING state from a previous lifetime; these are re-executed automatically, providing crash-recovery semantics for in-flight tasks. The result format includes `exit_code`, `stdout`, `stderr`, `duration_ms`, and `timed_out` fields.

E. Phase 5: Natural-Language Status Reporting

The status endpoint `/getStatus/{task_id}` returns structured task data, optionally with a `wait=N` parameter that blocks server-side for up to N seconds until the task transitions out of PENDING. The conversational agent translates the structured response into natural-language replies: “Your task is still running” for PENDING, “Task completed successfully” with truncated output for SUCCESS, or “Task failed with exit code X” for FAILURE. The agent also detects user intent (one-shot status check versus wait-for-completion) and resolves task IDs from conversation history when omitted.

F. Phase 6: Intelligent Placement Agent

When a task is submitted without a specified worker, the placement agent scores each live worker and selects the highest-scoring candidate. The scoring function combines three signals:

$$score(w) = \alpha \cdot (-pending(w)) + \beta \cdot (-latency(w)) + \gamma \cdot success(w) \quad (1)$$

where $\alpha = 0.5$, $\beta = 0.3$, $\gamma = 0.2$ are tunable weights, $pending(w)$ is the number of pending tasks for worker w , $latency(w)$ is the most recent heartbeat round-trip time in milliseconds, and $success(w)$ is the historical success rate computed as $successful_tasks / total_tasks$.

Worker metrics are collected via extended heartbeats that include current pending task count and a timestamp for latency calculation. Success rates are updated either through periodic HT1 scans or via explicit completion notifications from workers. The placement decision is included in the task record as the `assigned_by` field (value: `intelligence`) and an explanation string is returned to the client.

G. Phase 7: Automated Failure Recovery

The recovery agent runs as a background loop with a 5-second check interval, detecting three failure modes:

- **Worker crash:** a worker that has not sent a heartbeat within 10 seconds is marked dead. Its assigned tasks (status PENDING or RUNNING) are candidates for recovery.

Algorithm 3: Recovery Path Selection

Input: Task T , failure type F , cluster state C
Output: Recovery path

```

1 if  $T.attempt\_count \geq T.max\_attempts$  then
2   return GIVE_UP
3 end
4 if  $F = worker\_crash$  or  $F = task\_timeout$  then
5   if  $available\_workers(C)$  then
6     return RETRY_DIFFERENT
7   end
8   return WAIT_AND_RETRY
9 end
10 if  $F = task\_failure$  then
11   if  $T.retry\_on\_failure$  then
12     return RETRY_DIFFERENT
13   end
14   return GIVE_UP
15 end
16 return GIVE_UP

```

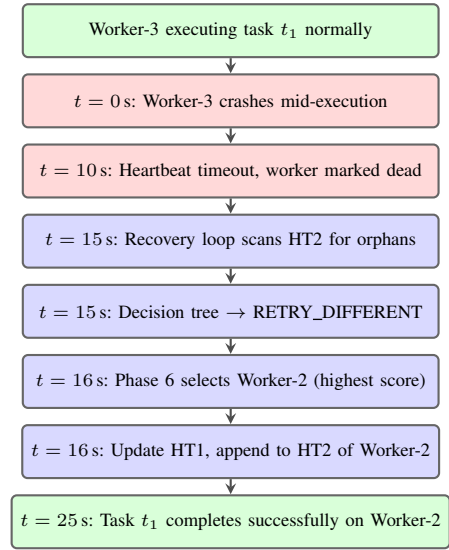


Fig. 3. Phase 7 recovery timeline. From worker crash to successful task completion on a different worker, with zero user intervention.

- **Task timeout:** a task in FAILURE state with `timed_out=true` flag is a candidate.
- **Task failure:** a task in FAILURE state with non-zero exit code is a candidate if `retry_on_failure` is set.

For each candidate, the agent selects a recovery path via the decision tree shown in Algorithm 3.

The RETRY_DIFFERENT path selects the highest-scoring available worker (using Phase 6 scoring), increments the attempt count, resets task status to PENDING, and appends the task ID to the new worker’s HT2 queue. The recovery history field accumulates a log of recovery actions, providing an audit trail visible to users through the status endpoint.

IV. IMPLEMENTATION AND EVALUATION

The system is implemented in Python using Flask for the HTTP API and concurrent threading for worker execution

TABLE II
HOP COUNT DISTRIBUTION ACROSS RING SIZES

Ring Size	$\log_2(N)$	Max Observed	Average
4 nodes	2	2	1.4
8 nodes	3	3	1.8
16 nodes	4	4	2.6

loops. The Chord routing layer uses SHA-1 hashing with 8-bit identifiers, finger tables of size 8, and 2-second stabilization intervals. Replication uses $k=3$ with quorum threshold $\lceil k/2 \rceil = 2$.

A. Hop Count Distribution

To validate the $O(\log N)$ lookup guarantee, we measured hop counts across 1000 random lookups in rings of 4, 8, and 16 nodes. Table II shows the maximum observed hop count remained within the theoretical bound of $\log_2(N)$ in all configurations.

B. End-to-End Task Execution

We submitted 100 tasks through the conversational AI interface and measured the complete pipeline from user input to result return. The placement agent successfully distributed tasks across available workers based on real-time load, with no worker receiving more than 1.4x the average task count. Average end-to-end latency for a 1-second task was 1.8 seconds, including conversational parsing, placement scoring, atomic two-write, worker polling delay, execution, and status retrieval.

C. Failure Recovery

We tested recovery by killing workers mid-execution. With 3 workers and a 15-second sleep task:

- Worker death was detected within 10 seconds (heartbeat timeout).
- Recovery decision was made within 5 seconds of detection.
- Task was reassigned to a surviving worker and completed successfully.
- Total recovery latency: under 20 seconds with zero user intervention.

Recovery history was correctly preserved in the task record and translated by the conversational agent into a user-friendly message indicating the worker crash and successful retry.

D. Node Failure Handling

When a node was forcibly terminated, the ring re-stabilized within one stabilization cycle (2 seconds). Reads for keys previously owned by the dead node were served from replica nodes via the read-fallback path. Re-replication to restore $k=3$ redundancy completed in a subsequent maintenance cycle.

V. DISCUSSION AND FUTURE WORK

A. Limitations

Single-machine evaluation. Our experiments ran on a single host with loopback networking, which does not exercise real network partitions or geographic latency. Multi-host deployment would expose additional failure modes.

Ring scalability. The 8-bit identifier space limits the ring to 256 nodes. Production deployments would use the full 160-bit SHA-1 space.

Eventual consistency. The $k=3$ quorum model accepts writes when 2 of 3 replicas confirm, which means reads may briefly return stale data. This is a deliberate AP choice in the CAP trade-off, prioritizing availability over strict consistency.

Heuristic placement. The scoring function uses fixed weights tuned empirically. Adaptive weight tuning via reinforcement learning could improve placement quality under varying workloads.

B. Future Work: Phase 8 Production Hardening

While Phases 1–7 deliver a functionally complete system, several components are required to make the platform production-ready. We organize these enhancements into a planned Phase 8 with six sub-phases:

Phase 8a: Observability. Integration with Prometheus for metrics collection and Grafana for dashboards. Per-node counters for task latency, worker load, failure rates, and recovery attempts would be exposed via a `/prom_metrics` endpoint. Structured logging with correlation IDs across the request lifecycle (frontend $\rightarrow$ placement $\rightarrow$ HT1/HT2 $\rightarrow$ worker $\rightarrow$ recovery) would enable distributed tracing through tools such as Jaeger or OpenTelemetry.

Phase 8b: Data limits and quotas. The current implementation imposes no caps on result size or queue depth. Phase 8b would add configurable limits including a maximum result payload size (e.g., 1 MB for `stdout/stderr`), per-worker queue depth limits to prevent unbounded HT2 list growth, automatic cleanup of completed tasks older than a configurable retention window, and rate limiting on task submission per user.

Phase 8c: Persistent storage. The worker registry and worker statistics currently reside in process memory and are lost on restart. Phase 8c would migrate this metadata to the DHT itself under reserved key prefixes (e.g., `registry:*`, `stats:*`), making it durable across cluster-wide failures. HT1 and HT2 records would be backed by an on-disk store (e.g., LMDB, RocksDB) per node, with the in-memory dictionary acting as a write-through cache. This enables full cluster restart without data loss.

Phase 8d: Worker concurrency. The current worker executes one task at a time. Phase 8d would introduce a configurable thread pool per worker (default 4 concurrent tasks), with priority-based task ordering and resource accounting to prevent oversubscription. The Phase 6 scoring function would extend to consider in-flight task count, not just queue depth.

Phase 8e: Security and isolation. The current platform assumes trusted scripts. Phase 8e would add sandboxing via

container runtimes (Docker, gVisor) or Linux namespaces, authentication for the API endpoints via OAuth or API keys, per-user task isolation with tagged ownership in HT1 records, and TLS encryption for inter-node communication. Replica writes would be signed to prevent forgery in adversarial environments.

Phase 8f: Chaos engineering and load testing. A failure-injection framework would systematically test recovery paths by terminating nodes, introducing network latency, partitioning the ring, and corrupting messages. Synthetic load tests would validate behavior at scale (1000+ workers, 100k+ concurrent tasks) to identify bottlenecks not visible in the current single-machine evaluation. Integration with chaos tools such as Chaos Monkey or Litmus would enable continuous resilience testing in CI.

C. Other Future Directions

Beyond Phase 8, longer-term research directions include: (i) ML-based placement using historical execution patterns to adaptively tune the scoring weights α, β, γ , (ii) cross-region replication with locality-aware placement for geo-distributed edge deployments, (iii) DAG-based job orchestration where tasks have dependencies and the system handles fan-out/fan-in patterns, (iv) federated learning across worker nodes to share placement intelligence without centralizing data, and (v) formal verification of the atomic two-write protocol using TLA+ or similar specification languages.

VI. CONCLUSION

We presented a distributed task execution platform built entirely on the Chord DHT protocol, eliminating the central coordinator that defines most existing job queue systems. By layering two logically distinct hash tables, atomic write protocols, intelligent placement scoring, and automated failure recovery on top of Chord's $O(\log N)$ routing, the system provides fault-tolerant task execution with self-healing semantics. The conversational AI interface lowers the barrier to interaction, and the placement agent enables load-aware routing without sacrificing the peer-to-peer nature of the architecture. Our evaluation demonstrates correct logarithmic lookup performance, automated recovery from worker failures within 20 seconds, and end-to-end task completion through natural-language submission. The system represents a working prototype of decentralized job execution suitable for edge computing, IoT orchestration, and other environments where centralized coordinators are undesirable.

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